

CGVP-MedGemma: A Confidence-Gated Visual Prior Consistency Framework for Multimodal Skin Lesion Classification

Ruixi Zhang^a, Chen Xiong^{a,*}

^a*Sun Yat-sen University, School of Intelligent Systems Engineering, Shenzhen, 518000, China*

Abstract

Skin lesion classification is important for early diagnosis and better patient prognosis. Existing deep learning methods generally rely on feature extraction from lesion images and incorporate patients clinical information to improve diagnostic performance. However, these methods still suffer from insufficient representation learning, modality bias, and unstable predictions in clinical scenarios. To address these issues, we propose CGVP-MedGemma, a confidence-gated visual prior consistency framework based on a medical vision-language model. The framework combines lesion images with clinical descriptions as input to MedGemma-4B-IT. The model is then fine-tuned with LoRA, a parameter-efficient strategy that enables effective adaptation of the large vision-language model to the skin lesion domain with limited training samples. Meanwhile, the model's output is restricted to a pre-defined diagnostic label set, which converts open-ended generation into a closed-set classification problem and mitigates label hallucination. To mitigate the models over-reliance on textual metadata and its neglect of visual evidence, we construct an independent visual prior branch and introduce a KL-divergence-based consistency regularization term, which encourages multimodal predictions to remain grounded in reliable visual evidence. The proposed method achieves an accuracy of 0.798 on PAD-UFES-20 and 0.784 on HAM10000¹, providing a feasible approach for applying medical vision-

*Corresponding author

Email address: xiongch8@mail.sysu.edu.cn (Chen Xiong)

¹The code is available at <https://github.com/zhangrx59/Graduation-Project>.

language models to skin tumor-assisted diagnosis.

Keywords: Skin lesion classification, Medical vision-language model, Multimodal learning, Visual prior consistency, Parameter-efficient fine-tuning.

1. Introduction

Deep learning has significantly advanced skin lesion classification. Early studies mainly relied on convolutional neural networks (CNNs). To further improve diagnostic performance, subsequent studies incorporated residual connections, dense blocks, and attention mechanisms to improve feature extraction and classification performance[1, 2, 3].

Despite these advances, two critical challenges remain to be resolved. First, conventional methods mainly focus on visual representation learning but underutilize text information. Studies have shown that dermoscopic images and patient metadata are complementary, proper multimodal fusion can improve diagnostic performance[4, 5]. But traditional architectures often rely on task-specific integrated structures and lack the advantages of knowledge transfer and evidence-based interpretability offered by medical vision-language models (MVLMs) through modeling images and text within a unified framework. Second, when applied to small-scale medical classification tasks, MVLMs face challenges such as high training costs and output hallucinations[6, 7, 8].

To address these issues, recent studies have sought to optimize model accuracy through concept-adaptive fine-tuning or the creation of domain-specific datasets[6, 7]. However, existing methods still fall short in addressing class imbalance in clinical tasks and the shortcut-learning bias in multimodal fusion[9]. Models may rely too much on textual metadata while neglecting critical visual features, resulting in poor generalization in complex cases. Therefore, adapting generative MVLMs to closed-set classification tasks and mitigate modal bias while seamlessly integrating images with clinical information remains a key challenge.

To this end, this paper proposes a framework for fine-tuning MedGemma-4B-IT[8] for closed-set classification of skin lesions. This framework aims to address the issues of output instability and modal bias in generative large language models when applied to specialized medical diagnosis. By combining expert-level visual priors with parameter-efficient fine-tuning techniques[10],

we have developed a diagnostic model that preserves clinical semantic understanding while accurately capturing local pathological features, thereby providing an efficient solution for small-scale, class-imbalanced dermatology datasets.

The main contributions of this study are as follows:

- We propose a multimodal VLM adaptation framework for closed-set diagnosis. It converts structured clinical metadata into natural-language notes and constrains the output to a predefined label set, reformulating open-ended generation as closed-set softmax classification and eliminating label hallucination.
- We design a confidence-gated visual prior consistency mechanism that adopts an independent ResNet branch to provide auxiliary class distributions and applies KL-divergence regularization when the branch’s prediction confidence exceeds a threshold, mitigating textual shortcut bias without incurring additional inference overhead.
- We improve recognition of rare classes through a joint strategy of frequency class weighting and label smoothing. And we validated on HAM10000 and PAD-UFES-20[11, 12], demonstrate the effectiveness of the proposed framework.

2. Related Works

2.1. Skin lesion classification

2.2. Metadata fusion in dermatology

2.3. Medical vision-language models

2.4. Class imbalance and shortcut learning

3. Method

3.1. Problem Formulation

In this study, we formulate skin lesion diagnosis as a multimodal classification task. For each sample, the input consists of a skin lesion image x and its metadata c , where c comprises 22 structured attributes including personal information, medical history, and lesion information. The output is a diagnostic label y from a predefined label set $\mathcal{Y}$. Therefore, the task is to learn a mapping $f_\theta(x, c) \rightarrow y$, where $y \in \mathcal{Y}$ and θ denotes the model parameters. Accordingly, the model prediction is represented as the conditional

probability distribution $p_\theta(y \mid x, c)$, and the final prediction is obtained by $\hat{y} = \arg \max p_\theta(y \mid x, c)$.

Under this setting, the model jointly exploits lesion images and clinical metadata, where the former provide visual evidence and the latter offer complementary contextual cues. Our goal is to improve the discriminative ability and prediction stability of skin lesion classification in small-scale and class-imbalanced scenarios by jointly exploiting both modalities.

3.2. Multimodal VLM Adaptation

Rather than directly concatenating tabular feature vectors, we serialize all attributes in c into a natural language clinical note, which is combined with x under an instruction-following format as unified multimodal input.

In this study, we adopt MedGemma-4B-IT, whose visual encoder is pre-trained on large-scale medical imaging corpora, providing a strong initialization that is already sensitive to clinically relevant visual features. This choice substantially reduces the domain gap that fine-tuning must bridge, making adaptation more data-efficient.

Given the limited scale of available datasets, full fine-tuning of a 4B model risks catastrophic forgetting. Therefore, we adopt LoRA, which injects trainable low-rank decomposition matrices into the attention and MLP projections, representing each weight update as $\Delta \mathbf{W} = \mathbf{B}\mathbf{A}$, where $\mathbf{A} \in \mathbb{R}^{r \times d}$, $\mathbf{B} \in \mathbb{R}^{d \times r}$, and $r \ll d$. By constraining adaptation to a low dimensional subspace, LoRA preserves pretrained representations while directing the model’s capacity toward task-specific discrimination.

Because a VLM’s output space includes the entire vocabulary, standard next-token prediction treats diagnosis as a sequence generation problem rather than a K -way discrimination task. This not only introduces label hallucination but also results in poorly calibrated probability estimates, since softmax normalization is performed over tens of thousands of irrelevant tokens. To resolve this, we construct a closed-set classification head. At the answer token position, we extracting only the logits corresponding to the K label tokens and renormalize them within the label set:

$$p_\theta(y \mid x, c) = \frac{\exp(z_y)}{\sum_{y' \in \mathcal{Y}} \exp(z_{y'})} \quad (1)$$

The classification objective is a label-smoothed weighted cross-entropy

loss:

$$\mathcal{L}_{\text{cls}} = - \sum_{k=1}^K w_k \cdot \tilde{y}_k \log p_{\theta}(y_k \mid x, c) \quad (2)$$

where w_k is normalized from the inverse square root of each class frequency to alleviate class imbalance, and $\tilde{y}_k$ denotes the smoothed soft label. To prevent the model from exploiting fixed label positions as a spurious shortcut, the enumeration order of label candidates in the prompt is randomly shuffled at each training iteration.

3.3. Prior-Guided Consistency Learning

While the previous section introduced an adaptation method for multi-modal data fusion, a critical risk emerges in small-scale, class-imbalanced clinical scenarios: when clinical metadata is strongly correlated with a specific diagnosis, the model may preferentially rely on the metadata while underestimating the visual evidence in x .

To alleviate this issue, we introduce an independent visual prior branch. This branch adopts a ResNet50 as the backbone and embeds a Convolutional Block Attention Module (CBAM) [13] after the final residual stage. Channel attention and spatial attention mechanisms are applied to selectively enhance feature channels associated with lesion recognition patterns and effectively guide the model’s receptive field toward lesion regions. The visual prior branch is trained with class-balanced Focal Loss[14] combined with a weighted random sampler to handle long-tailed class distributions. After training, its parameters remain frozen during the LoRA fine-tuning of the main model. Crucially, this branch takes only x as input and has no access to c , ensuring that its output distribution constitutes a pure visual diagnostic reference, structurally decoupled from any form of textual shortcut.

During LoRA fine-tuning, the frozen visual prior branch performs a forward pass on each training image to produce a purely visual reference distribution $p_r(y \mid x)$. This distribution serves solely as a consistency target within the loss function and is never injected into the prompt. We apply a KL divergence regularization term to penalize excessive deviation of the main model’s output from this visual reference, ensuring that multimodal predictions remain grounded in visual evidence even when textual cues are strongly predictive. Specifically, we define the set of high-confidence samples

as:

$$\mathcal{G} = \{ i \mid \max_{y \in \mathcal{Y}} p_r(y \mid x_i, c_i) > \tau \}, \quad (3)$$

where τ is a confidence threshold. The consistency loss is then computed as:

$$\mathcal{L}_{\text{KL}} = \frac{1}{|\mathcal{G}| + \epsilon} \sum_{i \in \mathcal{G}} D_{\text{KL}}(p_{\theta}(y \mid x_i, c_i) \parallel p_r(y \mid x_i, c_i)), \quad (4)$$

where ϵ is a constant for numerical stability, D_{KL} measures the discrepancy between the output distribution of the main model and the reference distribution produced by the auxiliary prior branch. In this way, the main model is encouraged to preserve multimodal discrimination ability while avoiding excessive deviation from reliable prior evidence. The overall training objective is defined as:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda \mathcal{L}_{\text{KL}} \quad (5)$$

where $\mathcal{L}_{\text{cls}}$ denotes the main classification loss and λ controls the strength of consistency regularization. Since the auxiliary prior branch is used only during training, the proposed mechanism introduces no additional inference time overhead.

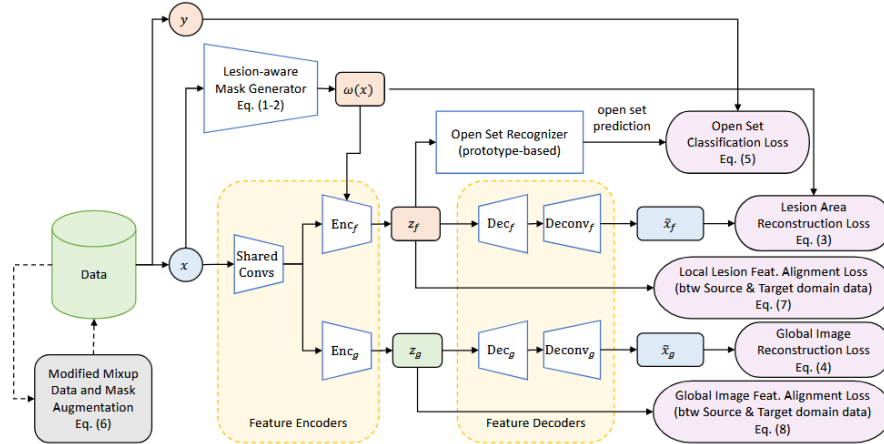


Figure 1: Technological route.

3.4. Training Strategy

The visual prior branch adopts an ImageNet-pretrained ResNet50 with an input resolution of 320×320 . To mitigate class imbalance, we employ

weighted random sampling and Focal Loss ($\gamma = 1.5$), optimizing the model with AdamW and a ReduceLROnPlateau scheduler. This branch is trained for 30 epochs, and then its parameters are frozen for subsequent fine-tuning.

For the main model, we apply LoRA fine-tuning using bfloat16 precision (rank = 32, $\alpha = 64$, dropout = 0.05). Training lasts for 15 epochs with a global batch size of 4. We adopt a cosine learning rate schedule with warmup (initial 5×10^{-5} , warmup ratio 0.1) and set the gradient clipping threshold to 0.5. The loss is computed only at the answer token positions. The regularization weight λ and the confidence gate threshold τ are set to 0.05 and 0.85, respectively, which are determined through ablation studies on the validation set.

4. Experiments

4.1. Datasets and Experimental Setup

Our experiments are conducted on two public skin lesion datasets, PAD-UFES-20 and HAM10000 [11, 12]. PAD-UFES-20 contains smartphone-captured clinical images and structured patient metadata across six skin lesion classes: actinic keratosis (ACK), basal cell carcinoma (BCC), melanoma (MEL), melanocytic nevus (NEV), squamous cell carcinoma (SCC), and seborrheic keratosis (SEK). HAM10000 consists of 10015 dermoscopic images covering seven diagnostic categories. To prevent data leakage, we split PAD-UFES-20 at the **patient level** into training (70%), validation (15%), and test (15%) sets. To preserve the original clinical records, we perform **no imputation or removal** for missing data, nor did we employ neural networks for missing value prediction, as such artificial imputation would introduce potential bias[15, 16, 17]. For cross-dataset evaluation, HAM10000 is used only on the shared label space with PAD-UFES-20 after category mapping, and non-overlapping categories are excluded from external evaluation.

To evaluate the model performance, we employ Accuracy (ACC), Recall, and Balanced accuracy (BACC). Let M denote the number of classes, N the total number of samples. For the i -th sample, let $\hat{y}_i$ and y_i be the predicted label and the ground-truth label. The metrics are defined as follows:

$$\begin{aligned}
\text{Accuracy} &= \frac{1}{N} \sum_{i=1}^N \mathbf{1}(\hat{y}_i = y_i) \\
\text{Recall}_j &= \frac{TP_j}{TP_j + FN_j} \\
\text{Balanced Accuracy} &= \frac{1}{M} \sum_{j=1}^M \text{Recall}_j
\end{aligned} \tag{6}$$

where TP_j and FN_j denote the true positives and false negatives for class j , respectively. Our baselines consist of three groups: **image-only**[18, 19, 20, 21, 22], **multimodal fusion**[4, 5, 23, 24, 25], and **open-source VLMs**[26].

Table 1: Performance comparison of five backbones combined with six metadata fusion methods on PAD-UFES-20.

Model	VGG13[18]		ResNet50[19]		EfficientNet-B4[20]		MobileNet-V2[21]		DenseNet-121[22]	
Method	ACC	BACC	ACC	BACC	ACC	BACC	ACC	BACC	ACC	BACC
Image-only	0.662	0.628	0.708	0.657	0.678	0.643	0.669	0.649	0.680	0.665
Concat[4]	0.701	0.713	0.732	0.725	0.735	0.708	0.712	0.717	0.722	0.720
MetaNet[23]	0.726	0.719	0.721	0.707	0.729	0.705	0.708	0.703	0.724	0.727
MetaBlock[24]	0.728	0.736	0.735	0.738	0.748	0.770	0.724	0.731	0.723	0.746
AuxNet[25]	0.751	0.748	0.757	0.749	0.743	0.728	0.736	0.711	0.748	0.739
DualRefNet[5]	0.753	0.745	0.759	0.748	0.751	0.738	0.746	0.724	0.756	0.749

4.2. Visual Prior Configuration

To determine the optimal visual prior branch and compare with fine-tuned VLMs, we evaluated both image-only and multimodal fusion methods on PAD-UFES-20. The results are presented in **Table 1**. Under the image-only setting, class imbalance significantly affected the performance of all networks, creating a notable gap between ACC and BACC. After incorporating metadata fusion, all networks achieved performance improvements. However, the visual prior branch in our framework serves as an auxiliary reference based on images. Since ResNet50 achieved the best overall performance under image-only setting, we adopt it as the backbone and further enhance it with CBAM to provide a reliable visual reference distribution for the subsequent LoRA fine-tuning stage.

4.3. Comparison With Other Models

We review recent studies and present the performance of representative VLMs on HAM10000 and PAD-UFES-20 in Table 2.

Table 2: Performance comparison of representative VLMs on HAM10000 and PAD-UFES-20.

Methods \ Dataset	HAM10000		PAD-UFES-20	
	ACC	Recall	ACC	Recall
LLaVA-Med-7B [27]	0.784	0.621	0.735	0.624
SkinVL-PubMM [7]	0.790	0.623	0.752	0.640
DermLIP [28]	0.881	–	0.740	–
Ours	0.784	0.656	0.798	0.776

Note: ‘–’ indicates the result is not reported in the original paper.

Our method shows strong performance on PAD-UFES-20 and remains competitive on HAM10000 compared with recent representative VLM baselines. These results suggest that the proposed framework remains competitive with conventional metadata fusion methods, while maintaining solid performance on the established HAM10000 benchmark.

4.4. Ablation Studies

We conduct an incremental ablation study on PAD-UFES-20 to quantify the contribution of each component in the proposed training strategy.

Table 3: Step-wise ablation of the training strategy on PAD-UFES-20.

Training Strategy	ACC	BACC
MedGemma-4B-IT	0.551	0.476
+ LoRA fine-tuning	0.630	0.608
+ Weighted CE loss	0.645	0.617
+ Visual prior branch	0.751	0.738
+ Confidence-gated KL	0.798	0.776

As shown in **Table 3**, the MedGemma-4B-IT baseline performs poorly, highlighting the challenge of applying generative VLMs directly to skin lesion diagnosis. LoRA fine-tuning substantially improves performance, demonstrating that lightweight domain adaptation aligns the pretrained model with the dermatology classification task. Adding weighted cross-entropy loss yields a smaller but consistent improvement, confirming that addressing class imbalance benefits the model, especially for underrepresented categories. A

substantial gain is observed after incorporating the frozen visual prior branch, indicating that it breaks the model’s over-reliance on metadata, stabilizing optimization and improving class-level balance. Finally, confidence-gated KL regularization further improves both ACC and BACC, achieving the best overall performance. This confirms that selective regularization on high-confidence prior predictions is most effective, reducing the risk of enforcing unreliable guidance on visually ambiguous samples.

Based on the final model, we quantified the contribution of clinical metadata by evaluating classification performance under different configurations. Metadata A contains **basic patient and symptom information**, while Metadata B extends Metadata A by **adding lesion morphological information**. As shown in **Table 4**, richer metadata improves classification performance, confirming that clinical attributes provide useful complementary information for accurate diagnosis.

Table 4: Performance of different metadata on PAD-UFES-20.

Metadata Variant	ACC	BACC
Metadata A	0.710	0.698
Metadata B	0.752	0.747
Full Metadata	0.798	0.776

5. Conclusion

In this paper, we proposed a multimodal fine-tuning framework for closed-set skin lesion diagnosis based on MedGemma-4B-IT. We convert structured metadata into natural-language notes with constrained output to a predefined label set, eliminating label hallucination. We further introduce a confidence-gated visual prior consistency mechanism in which a frozen ResNet50+CBAM branch regularizes predictions via KL divergence, mitigating textual shortcut bias without inference overhead. Experiments on PAD-UFES-20 show that each component contributes incrementally. Our framework achieves competitive results compared to representative VLMs and multimodal baselines. Beyond performance, our framework leverages VLMs’ generative nature to provide diagnostic reasoning, promoting trustworthy AI and evidence-based dermatology.

Data availability

The data used in this study are publicly available from the original sources cited in this article. PAD-UFES-20 and HAM10000 can be accessed through the resources described by Pacheco et al. [12] and Tschandl et al. [11]. No new dataset was generated for this study.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] L. Yu, H. Chen, Q. Dou, J. Qin, P.-A. Heng, Automated melanoma recognition in dermoscopy images via very deep residual networks, *IEEE Transactions on Medical Imaging* 36 (4) (2016) 994–1004.
- [2] K. M. Hosny, M. A. Kassem, Refined residual deep convolutional network for skin lesion classification, *Journal of Digital Imaging* 35 (2) (2022) 258–280.
- [3] J. Wu, W. Hu, Y. Wen, W. Tu, X. Liu, Skin lesion classification using densely connected convolutional networks with attention residual learning, *Sensors* 20 (24) (2020) 7080.
- [4] A. G. Pacheco, R. A. Krohling, The impact of patient clinical information on automated skin cancer detection, *Computers in Biology and Medicine* 116 (2020) 103545.
- [5] M. Khurshid, R. Singh, M. Vatsa, Multimodal dual-stage feature refinement for robust skin lesion classification, *Scientific Reports* 15 (1) (2025) 37775.

- [6] Y. Zhu, X. Wang, X. Huang, M. Xia, P. Mu, H. Hu, X. Zhang, Enhancing the interpretation of skin lesion diagnosis: Concept adaptive fine-tuning of vision-language models, *IEEE Journal of Biomedical and Health Informatics* (2025).
- [7] W. Zeng, Y. Sun, C. Ma, W. Tan, B. Yan, Mm-skin: Enhancing dermatology vision-language model with an image-text dataset derived from textbooks, in: *Proceedings of the 33rd ACM International Conference on Multimedia*, 2025, pp. 3769–3778.
- [8] A. Sellergren, S. Kazemzadeh, T. Jaroensri, A. Kiraly, M. Traverse, T. Kohlberger, S. Xu, F. Jamil, C. Hughes, C. Lau, et al., Medgemma technical report, arXiv preprint arXiv:2507.05201 (2025).
- [9] M. Watson, T. Winterbottom, T. Hudson, B. Jones, H. P. Shum, A. Atapour-Abarghouei, T. Breckon, J. Harmsworth King, N. Al Moubayed, Multimodal models for skin cancer classification using clinical freetext and dermatoscopic images, *Communications Medicine* (2026).
- [10] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, W. Chen, et al., Lora: Low-rank adaptation of large language models, *ICLR* 1 (2) (2022) 3.
- [11] P. Tschandl, C. Rosendahl, H. Kittler, The ham10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions, *Scientific Data* 5 (1) (2018) 180161.
- [12] A. G. Pacheco, G. R. Lima, A. S. Salomao, B. Krohling, I. P. Biral, G. G. De Angelo, F. C. Alves Jr, J. G. Esgario, A. C. Simora, P. B. Castro, et al., Pad-ufes-20: A skin lesion dataset composed of patient data and clinical images collected from smartphones, *Data in Brief* 32 (2020) 106221.
- [13] S. Woo, J. Park, J.-Y. Lee, I. S. Kweon, Cbam: Convolutional block attention module, in: *Proceedings of the European Conference on Computer Vision (ECCV)*, 2018, pp. 3–19.
- [14] T.-Y. Lin, P. Goyal, R. Girshick, K. He, P. Dollár, Focal loss for dense object detection, in: *Proceedings of the IEEE International Conference on Computer Vision*, 2017, pp. 2980–2988.

- [15] T. L. Vo, T. Nguyen, L. M. Lopez-Ramos, H. L. Hammer, M. A. Riegler, P. Halvorsen, Explainability of machine learning models under missing data, *Applied Soft Computing* (2026) 115105.
- [16] P. Ryan, B. Ozturk, L. Fearnley, T. Lawton, I. Habli, Does not impute! performance and ethical implications of missing data for an ai-based diabetes co-morbidity predictor, in: *International Conference on Computer Safety, Reliability, and Security*, Springer, 2025, pp. 511–523.
- [17] A. El Mir, E. B. de Sousa, I. Mesina-Estarrón, L. A. Celi, M. Hani, M. Benjelloun, N. Nageswaran, S. Mahmoudi, S. Siddiqui, S. Sadasivam, et al., Moving beyond the empty cell: The threat of decontextualized healthcare data, *PLOS Digital Health* 5 (1) (2026) e0001194.
- [18] K. Simonyan, A. Zisserman, Very deep convolutional networks for large-scale image recognition, *arXiv preprint arXiv:1409.1556* (2014).
- [19] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2016, pp. 770–778.
- [20] M. Tan, Q. Le, Efficientnet: Rethinking model scaling for convolutional neural networks, in: *International Conference on Machine Learning*, PMLR, 2019, pp. 6105–6114.
- [21] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, L.-C. Chen, Mobilenetv2: Inverted residuals and linear bottlenecks, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 4510–4520.
- [22] G. Huang, Z. Liu, L. Van Der Maaten, K. Q. Weinberger, Densely connected convolutional networks, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2017, pp. 4700–4708.
- [23] W. Li, J. Zhuang, R. Wang, J. Zhang, W.-S. Zheng, Fusing metadata and dermoscopy images for skin disease diagnosis, in: *2020 IEEE 17th International Symposium on Biomedical Imaging (ISBI)*, IEEE, 2020, pp. 1996–2000.
- [24] A. G. Pacheco, R. A. Krohling, An attention-based mechanism to combine images and metadata in deep learning models applied to skin cancer

- classification, *IEEE Journal of Biomedical and Health Informatics* 25 (9) (2021) 3554–3563.
- [25] M. Khurshid, M. Vatsa, R. Singh, Optimizing skin lesion classification via multimodal data and auxiliary task integration, in: *2024 IEEE International Symposium on Biomedical Imaging (ISBI)*, IEEE, 2024, pp. 1–5.
 - [26] S. Yan, Z. Yu, C. Primiero, C. Vico-Alonso, Z. Wang, L. Yang, P. Tschandl, M. Hu, L. Ju, G. Tan, et al., A multimodal vision foundation model for clinical dermatology, *Nature Medicine* 31 (8) (2025) 2691–2702.
 - [27] C. Li, C. Wong, S. Zhang, N. Usuyama, H. Liu, J. Yang, T. Naumann, H. Poon, J. Gao, Llava-med: Training a large language-and-vision assistant for biomedicine in one day, *arXiv preprint arXiv:2306.00890* (2023).
 - [28] S. Yan, M. Hu, Y. Jiang, X. Li, H. Fei, P. Tschandl, H. Kittler, Z. Ge, Derm1m: A million-scale vision-language dataset aligned with clinical ontology knowledge for dermatology, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025, pp. 12681–12690.